SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	2024 -CSE(AI)	**Date:**	20.10.26

Code of Conduct

I _T Bhanu Teja/192472259_ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____BhanuTeja____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Given:

- Total available spectrum = 45 MHz = 45,000 kHz
- Bandwidth required for each simplex channel = 25 kHz

Formula:

Bandwidth required for one full-duplex channel = 2 × Simplex channel bandwidth

= 2 × 25 = 50 kHz

Number of full-duplex channels = Total spectrum / Bandwidth per full-duplex channel

= 45,000 / 50 = 900

Answer: The telecom operator can support **900 full-duplex voice channels**, assuming the entire spectrum is available for pairing channels without additional guard bands.

Q2. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Given:

- Initial received power = 64 W
- Distance is doubled.
- Path-loss exponent (n) = 4

Formula:

$$P_2 = \frac{P_1}{(d_2/d_1)^n}$$

Since the distance is doubled:

$$P_2 = \frac{64}{2^4}$$

$$P_2 = \frac{64}{16} = 4 \text{ W}$$

Answer: The expected received power is **4 W**. When the distance doubles, the received power decreases by a factor of 16 because the path-loss exponent is 4.

Q3. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Given:

- Number of adjacent frequency bands = 30
- Width of each guard band = 40 kHz

Calculation:

The number of guard bands between 30 adjacent frequency bands is:

$$30 - 1 = 29$$

Total wasted bandwidth:

$$29 \times 40 = 1160 \text{ kHz}$$

Converting into MHz:

$$1160/1000 = 1.16 \text{ MHz}$$

Answer: The total bandwidth wasted due to guard bands is **1.16 MHz**. Guard bands help prevent interference between adjacent frequency bands and improve communication quality.

Q4. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Given:

- Total subscribers = 600,000
- Percentage of roaming subscribers = 15%

Formula:

$$\text{Roaming Subscribers} = \frac{15}{100} \times 600000$$
$$= 90{,}000$$

Answer: Approximately **90,000 subscribers** are expected to roam onto partner networks at any given moment, assuming the stated 15% applies to the current subscriber population. Roaming allows subscribers to access mobile services outside their home network through partner operators.

Q5. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Given:

- Transmitted power = 43 dBm
- Received power = −87 dBm
- Cell radius = 2 km
- Network frequency = 1800 MHz

Formula:

$$\text{Path Loss} = P_t - P_r$$

Where:

- P_t= Transmitted power
- P_r= Received power

Calculation:

$$\text{Path Loss} = 43 - (-87)$$

$$= 43 + 87 = 130 \text{ dB}$$

Answer: The signal experiences a **path loss of 130 dB** between the base station and the receiver at the cell edge. This loss occurs due to signal propagation over distance and environmental effects such as buildings, obstacles, and fading.